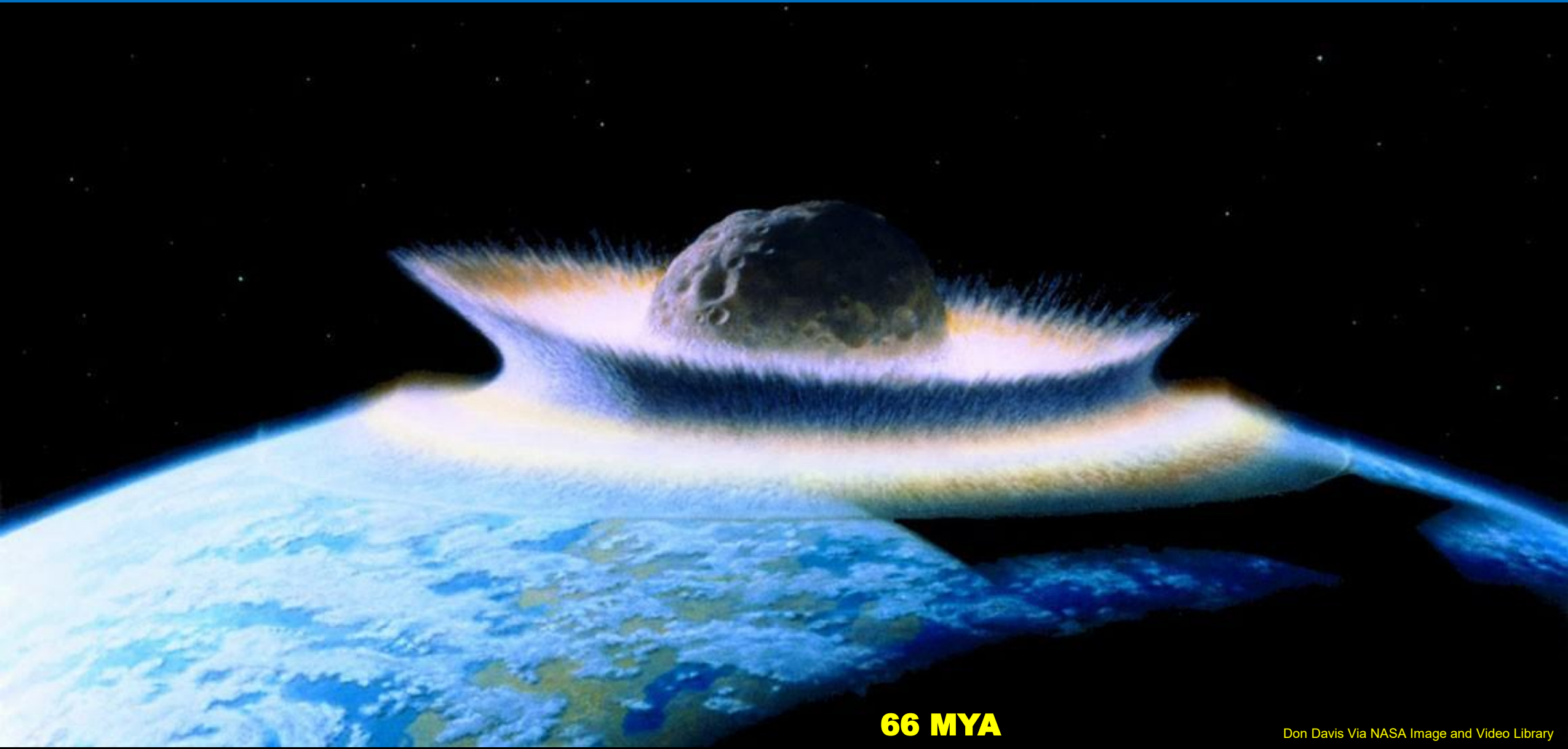


LECTURE 22: The End of The Dinosaurs

Chapter 17 pages 454 to 464; 474 to 485



66 MYA

Don Davis Via NASA Image and Video Library

LECTURE 22: The End of The Dinosaurs

Chapter 17 pages 454 to 464; 474 to 485

Outline

Part 1 – The K-Pg Extinction by M Ryan

**Part 2 – The End of the Cretaceous
(Dr. A. Alessandro Chiarenza)**

LECTURE 22: The End of The Dinosaurs

Chapter 17 pages 454 to 464; 474 to 485

Objectives

- **Learn about the Cretaceous-Paleogene boundary event**
- **Understand the factors of the Non-Avian Dinosaur Extinction**

What Killed Off The Dinosaurs?

- ? Asteroid Impact
- ? Deccan Traps
- ? Climate Change



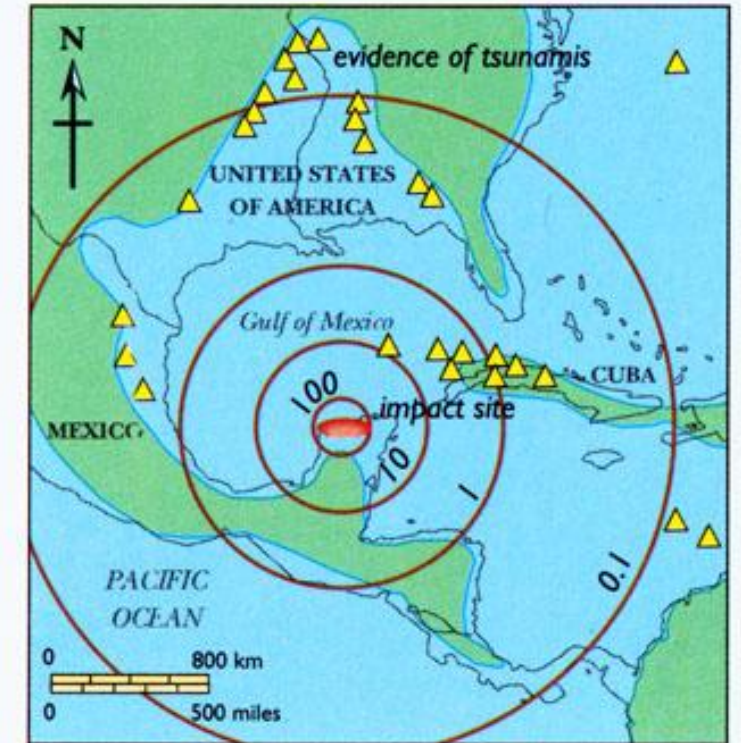
Cretaceous-Paleogene (K-Pg) Extinction Event



1980: **Luis & Walter Alvarez** discover an iridium anomaly in boundary layer. Impact hypothesis - an asteroid 10-15 km wide hit the Earth ~66 Ma!

1981: First evidence of 180 km wide Chicxulub crater in Gulf of Mexico in the Yucatan Peninsula presented by Glen Penfield

Boundary Clay ↗



I/Asteroids and iridium

10

lines of equal thickness of material ejected from crater

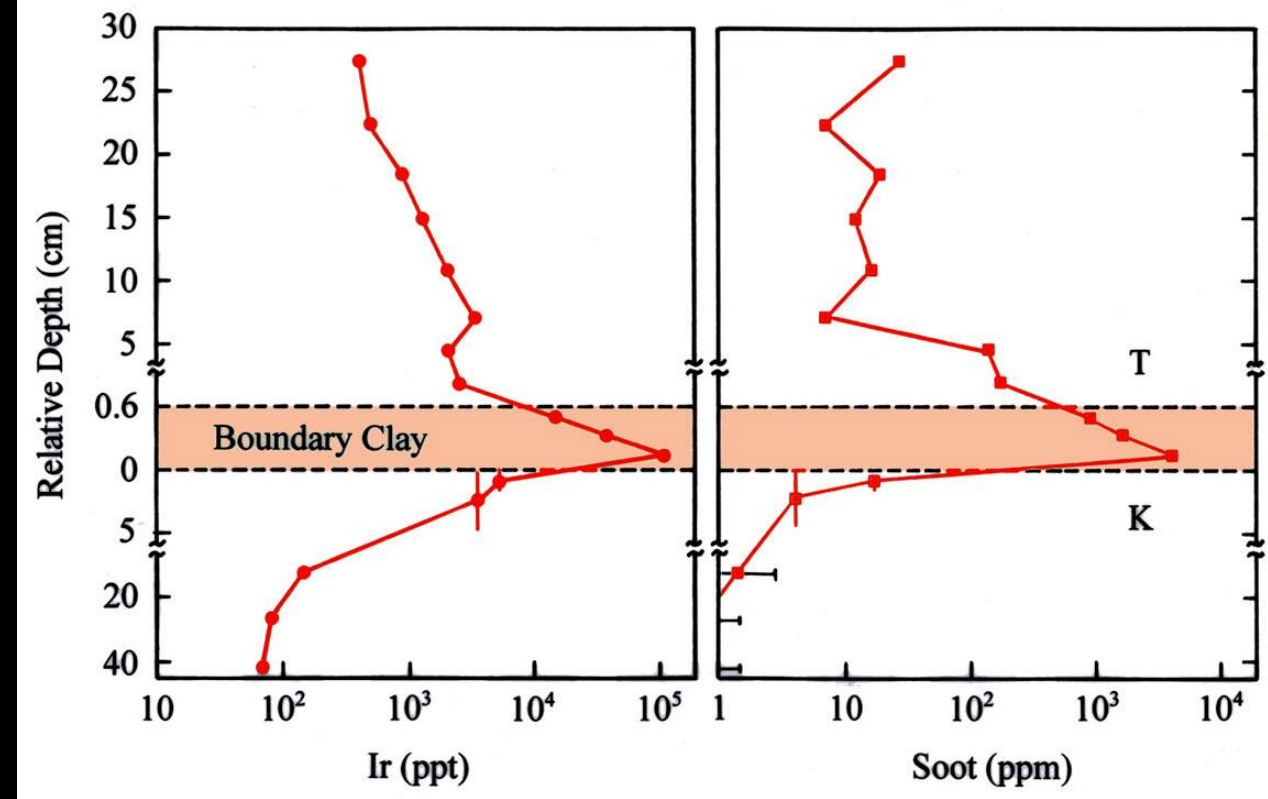
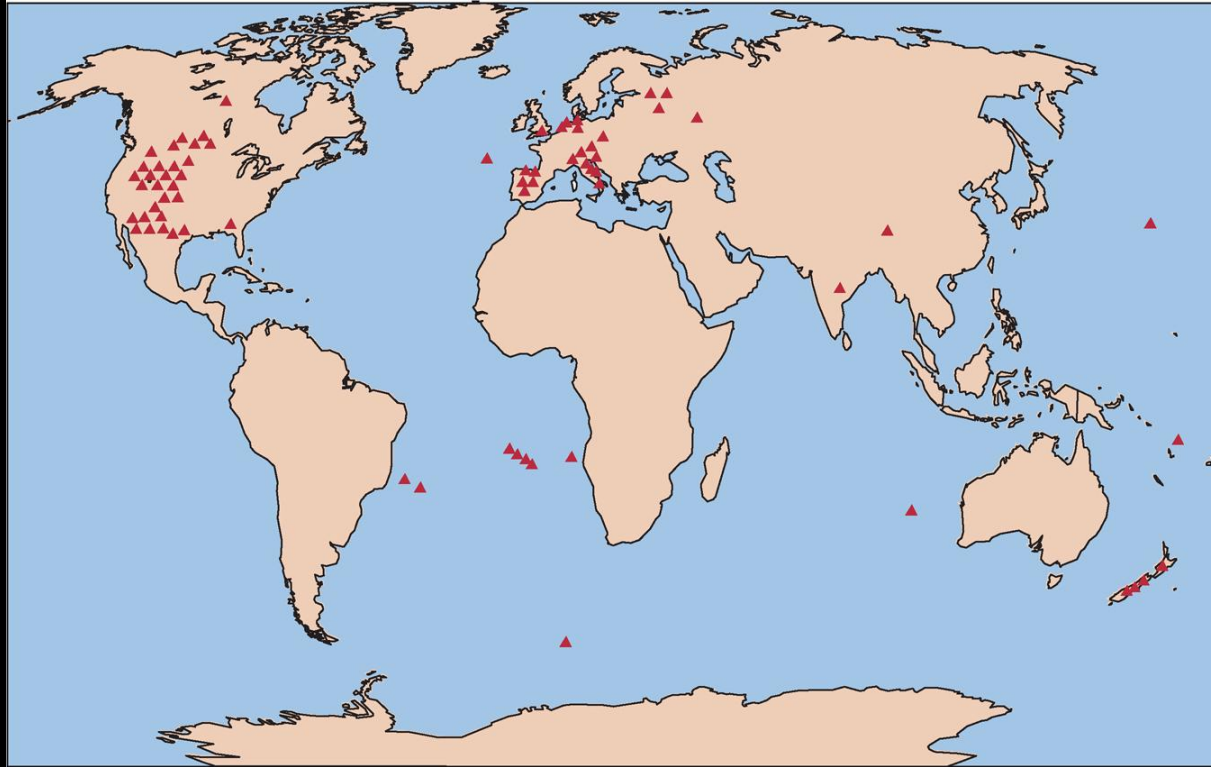


main iridium site

The Chicxulub impact site in the Yucatán peninsula.

The Iridium Anomaly

Before the impact crater was found, the Alvarez's predicted it. What was their hypothesis?



<https://www.lpi.usra.edu/science/kring/Chicxulub/global>

- Iridium (Ir) anomaly is found worldwide
- Present in boundary clay (BC) that is 1-20 cm thick depending on distance from impact site
- BC is the lithified fine-grained ash fallout from the impact
- Iridium (Ir) is ~0.3 ppb at Earth's surface; ~10 ppb in boundary clay

The Iridium Anomaly

- Extraterrestrial sources (e.g., asteroids) are enriched in Ir and other rare metals
- The distribution of BC sites in the world enriched with Ir predicted an ~ 10-15 km asteroid impact ~66 million years ago
- Presence of shocked quartz & microtektites in BC supported Impact Theory
- Discovery Chicxulub crater in 1981 & subsequent research validated the hypothesis

Shocked Quartz
Quartz that has been deformed due to intense pressure



Science Photo Library

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WN.87576



Microtektites
Rock melted & ejected at impact cool into silica-rich glass 'raindrops'. Their distribution can give clues to the locations of now buried craters

<http://www.tektites.co.uk/london-clay.html>

Cretaceous-Paleogene (K-Pg) Extinction Event

- ~10 to 15 km wide asteroid collided with Earth;
- The great cycles of nutrients that formed the complex food webs in the world's oceans temporarily shut down
- Many marine and terrestrial animals, as well as many plants, went extinct;
- Epic tsunamis rolled across ocean basins
- Landscapes were deforested;
- Wildfires likely raged.

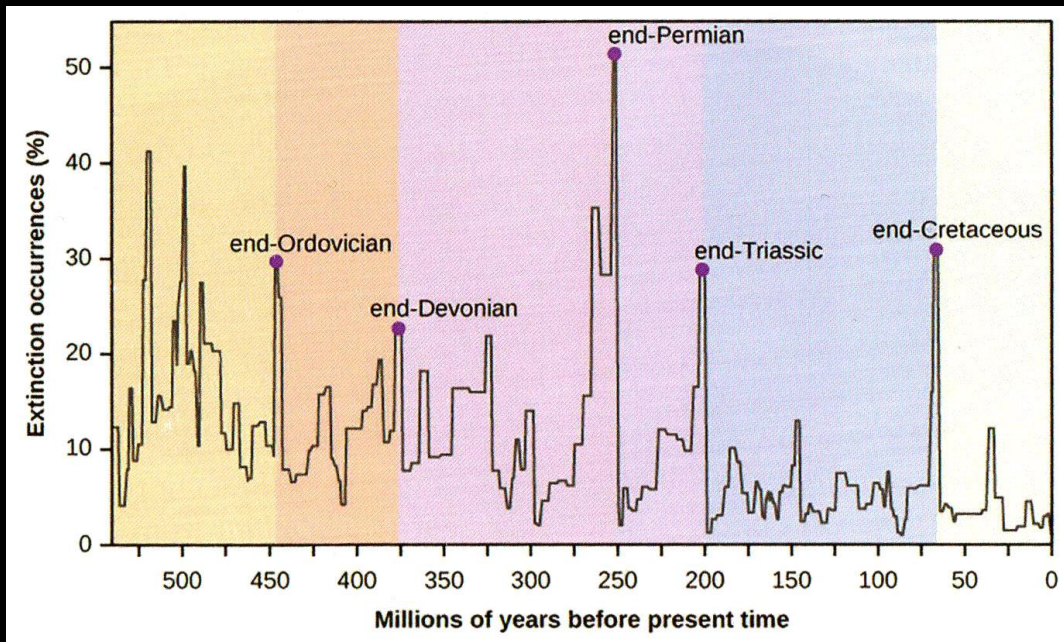


TABLE 20.1

EARTH'S FIVE LARGEST* EXTINCTION EVENTS IN CHRONOLOGICAL ORDER

Extinction	Age	% Extinct	Relative Size (based on species loss)
End Ordovician	444 Ma	86% of species 57% of genera	Second largest
Late Devonian	374-359 Ma	75% of species 35% of genera	Fifth largest
End Permian	252 Ma	96% of species 56% of genera	Largest
End Triassic	201 Ma	80% of species 47% of genera	Third largest
End Cretaceous	66 Ma	76% of species 40% of genera	Fourth largest

*These ranking vary on the methods, data, and counts (e.g., marine vs. total genera) used, see Bond and Grasby, 2017, for a comparison of different rankings. The End Permian always ranks at the top, however.

Cretaceous-Paleogene (K-Pg) Extinction Event



The Nadir Crater – reported in August 2022

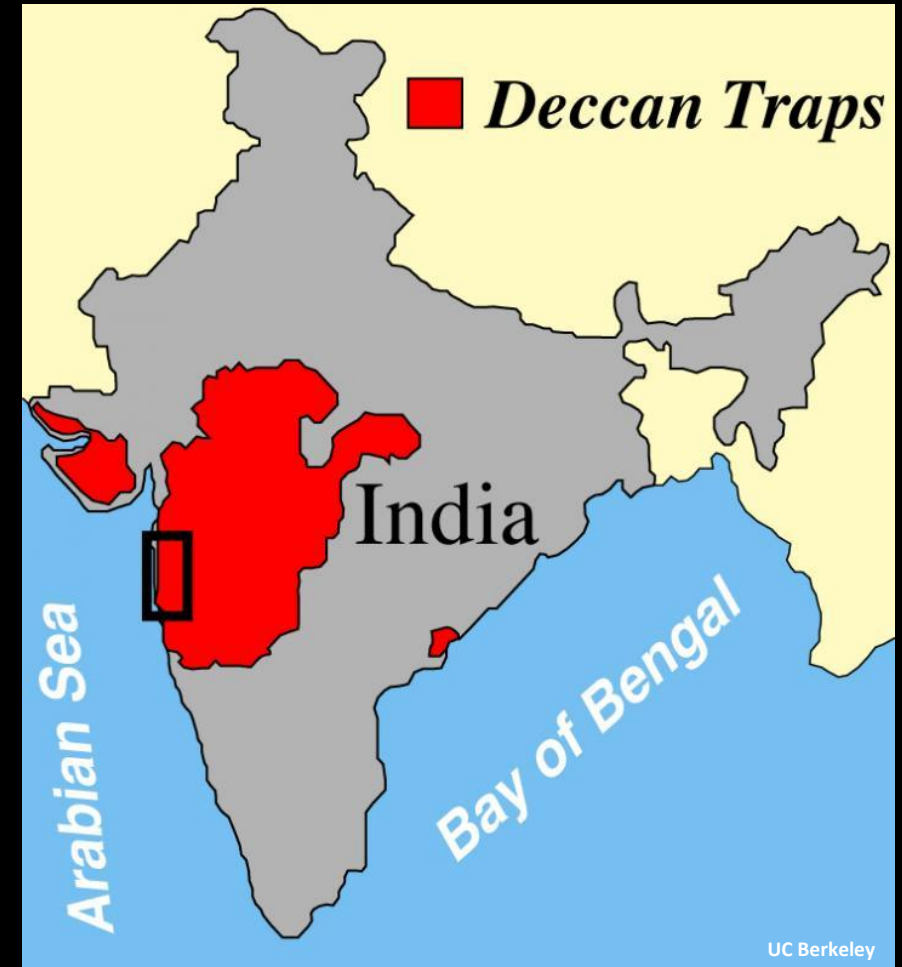
- 5 mile diameter crater buried 400 m off the coast of Guinea, West Africa
- Formed by a 400 m wide asteroid hitting 600 m of water
- ~ coincidental with Chicxulub crater
- Possibly part of the break up of a larger asteroid
- Resulted in a 100 m high tsunami & a 6.5 magnitude earthquake
- < 20 marine impact craters are known

Late Cretaceous Deccan Traps

- The Deccan Traps were a series of continental flood basalts: hot basaltic lava pouring through fissures in the Earth's crust
- May have lasted 30,000 years
- World's longest lava flows – 1500 km!
- Composed of three separate, closely timed events at the end of the Cretaceous
- Produced about >1.3 million km^3 of basalt
- Created a 3 km thick basaltic plateau in India at the K-Pg boundary
- Released CO_2 , SO_2 , NO_2 and methane, etc.

Did the Deccan Traps contribute to dinosaur extinction?

- New evidence suggests that they did not; rather they contributed to climatic recovery (warming)
(See video by Dr. Chiarenza)

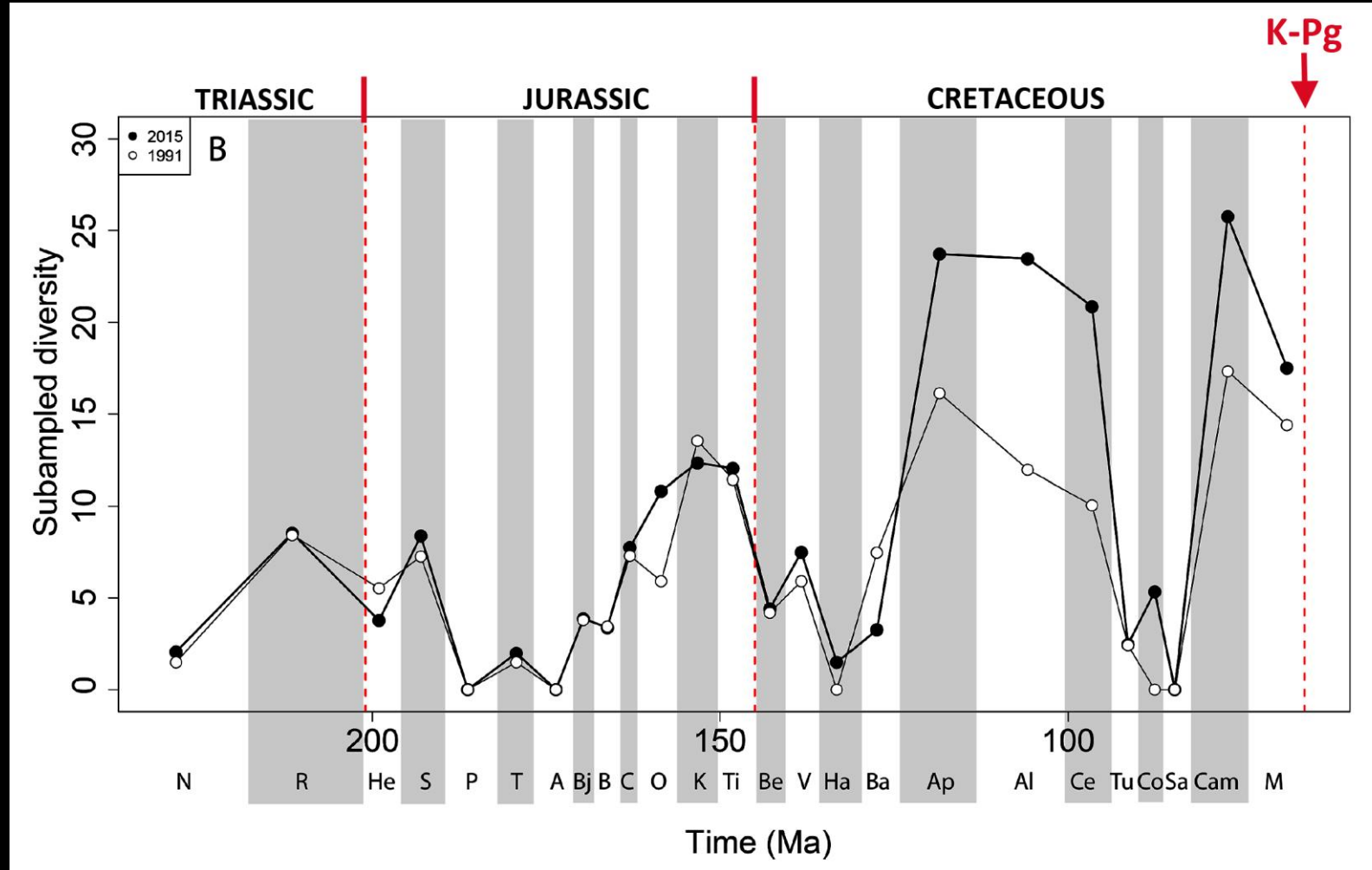


Late Cretaceous Deccan Traps



Sheikh, J. M. et al. 2020. Physical volcanology of the Pavagadh rhyolites, northern Deccan Traps: Stratigraphic, structural, and textural record of explosive and effusive eruptions. *Journal of Volcanology and Geothermal Research*, 107024. doi:10.1016/j.jvolgeores.2020.107024

Did Climate Change Kill Off The Dinosaurs?



Dinosaur diversity throughout the Mesozoic

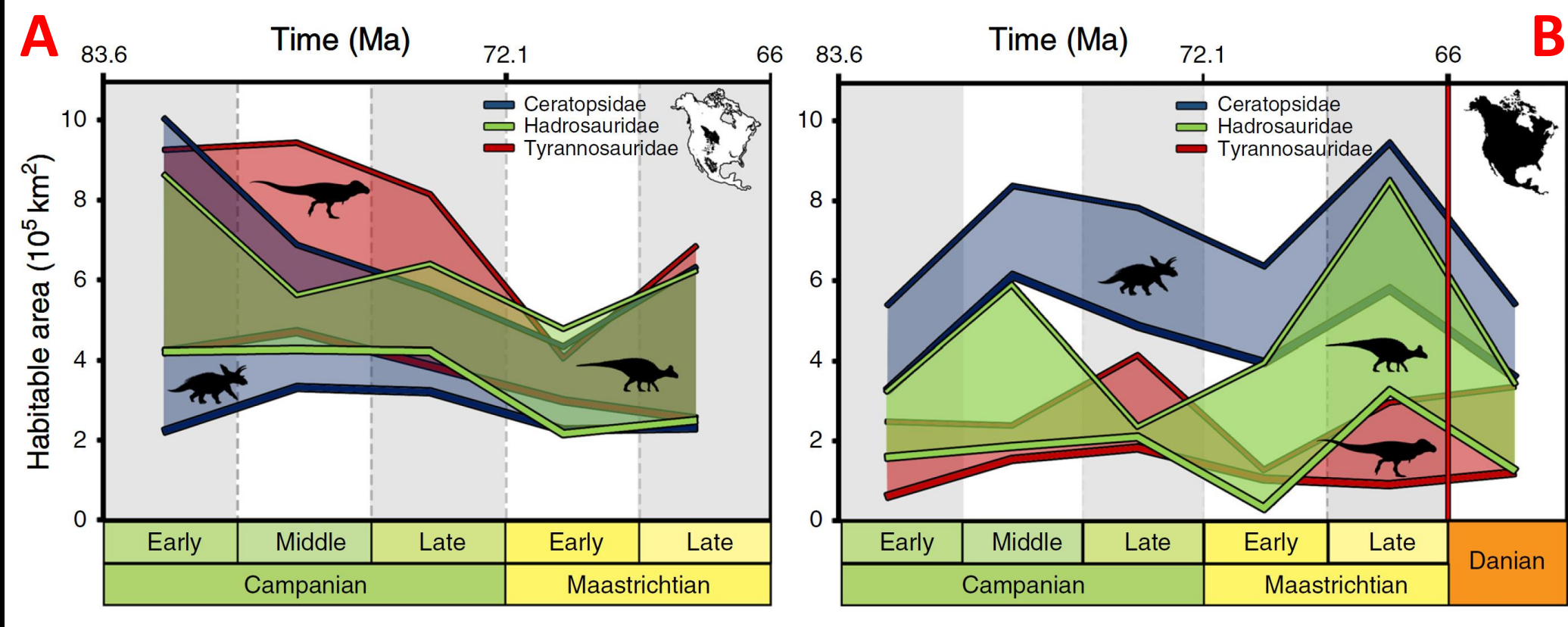


Figure: A. Alessandro Chiarenza

Prior to K-Pg Cretaceous/Paleogene extinction, dinosaur diversity is argued to either (1) have been in long-term decline, *or* (2) thriving until their sudden demise.

Ecological niche modeling shows **(A)** that Campanian-to-Maastrichtian (C to M) available habits decrease in area as C to M rock outcrop decreases (where fossils have been collected).

However, a continent-wide ecological model **(B)** shows habitat stability, or even a Camp-to-Maast. *increase* that is not preserved due to lack of outcrop to find fossils in.

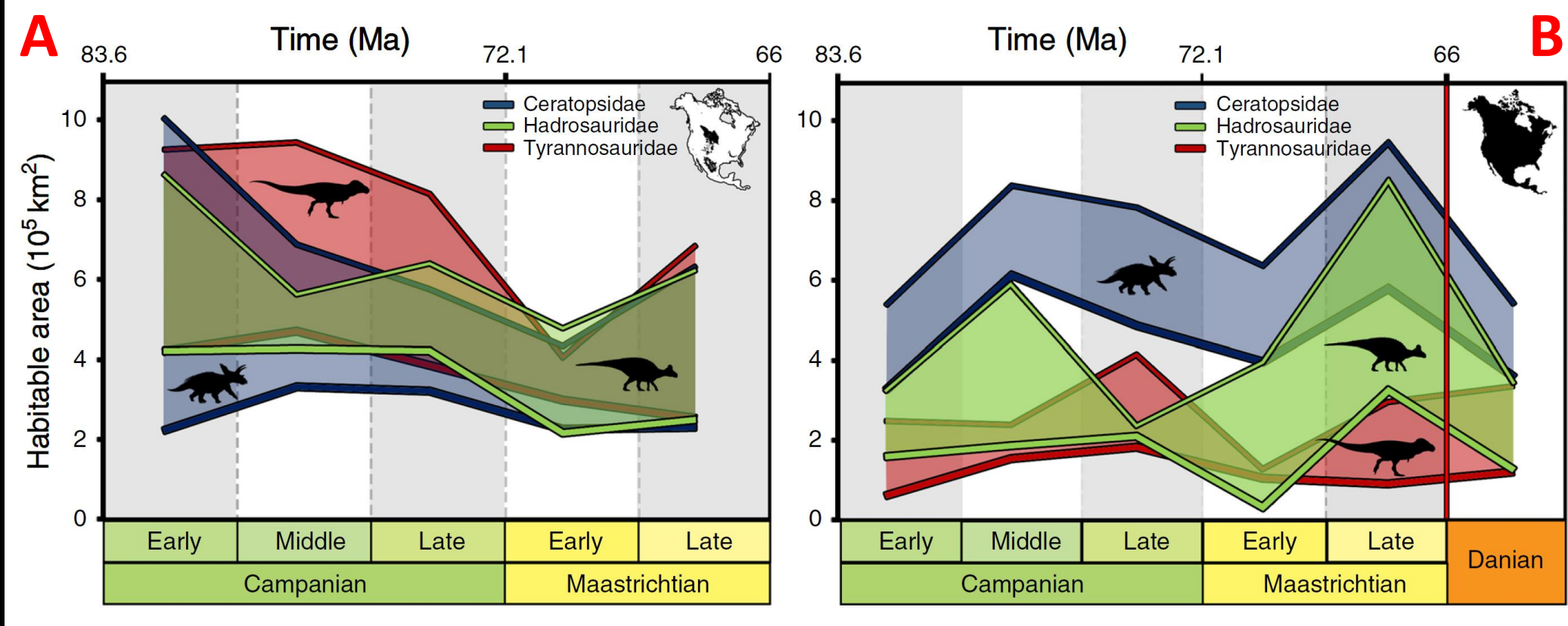


Figure: A. Alessandro Chiarenza

This loss of outcrop is due to the formation of the proto-Rocky Mountains and sea-level regression during the Maastrichtian.

Conclusion: The perceived loss of NA dinosaur diversity in the Maastrichtian is due to sampling bias, NOT due to a climatically-driven decrease in available habits

So, What Did Kill Off The Dinosaurs?

- X** Climate Change
- X** Deccan Traps
- ✓** Asteroid Impact

But, there is growing support for the idea both climate change and the Deccan Traps may have had at least a small contribution to the overall extinction event

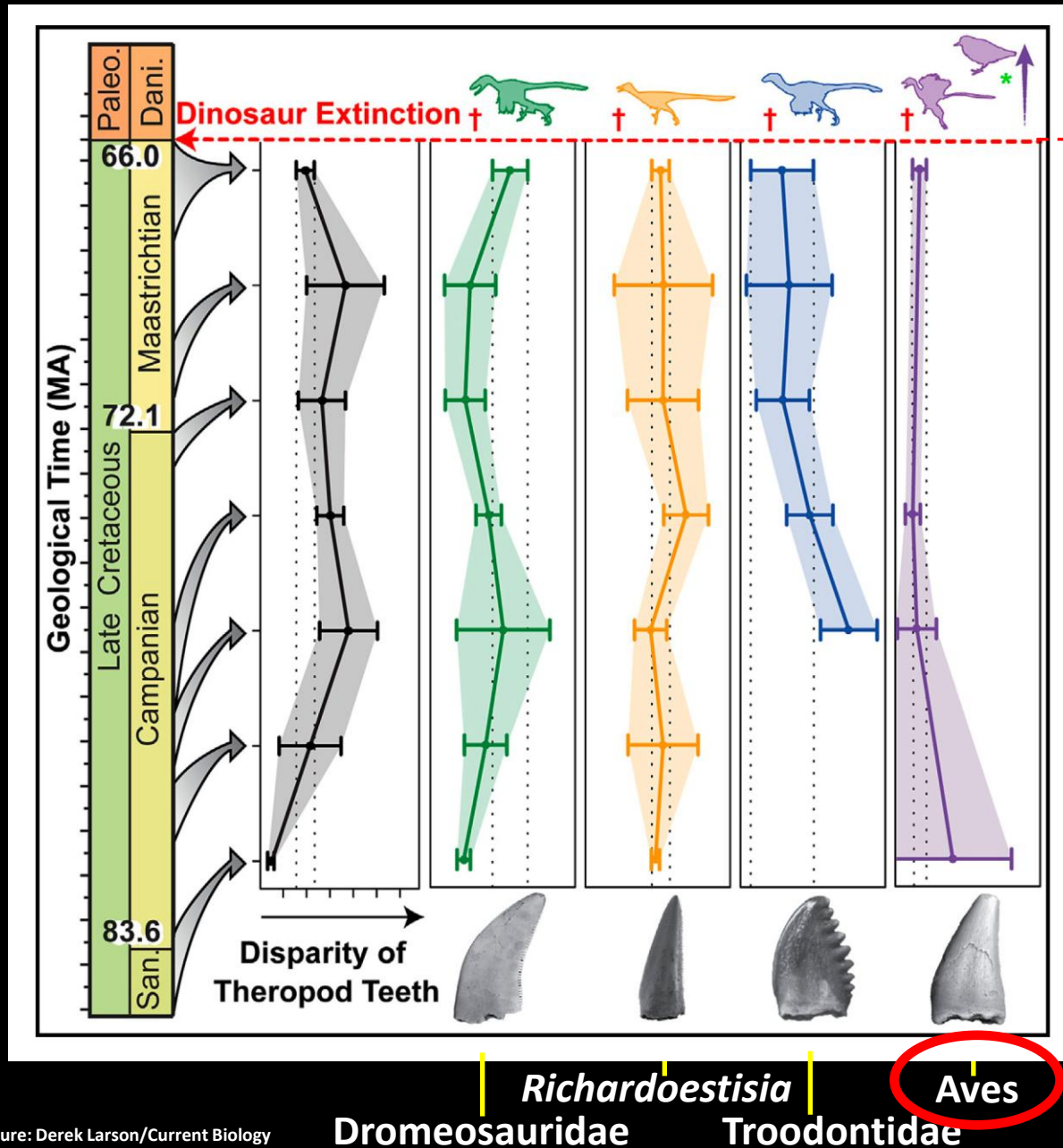


Why Did Birds Survive the K-Pg Extinction?



Art: Danielle Dufault

Why Did Birds Survive the K-Pg Extinction?



Birds with beaks survive

Aves with beaks appear in the latest Cretaceous

Toothed birds & maniraptorans with teeth

Many beaked birds are seed eaters

Post K-Pg extinction plant productivity was very low, but seeds can last for decades in the right conditions & may have been a good source of food during the Paleogene recovery

